

Database Project

AI-Powered Banking Compliance System

Schema Reference — 20 Tables

Reference / Lookup Tables

countries

Stores country risk ratings used for geography-based compliance rules.

Column	Type	Constraint	Notes	Description
country_code	CHAR(2)	PK, NOT NULL		ISO 3166-1 alpha-2 country code
country_name	VARCHAR(100)	NOT NULL, UNIQUE		Full country name
fatf_status	VARCHAR(20)	NOT NULL, DEFAULT 'whitelist'	CHECK: blacklist/greylist/whitelist	FATF risk classification
risk_score	DECIMAL(4,2)	NOT NULL, DEFAULT 0.0	CHECK: 0–10	Numeric risk score
is_sanctioned	BOOLEAN	DEFAULT FALSE		Whether country is sanctioned
updated_at	TIMESTAMP	DEFAULT NOW()		Last update timestamp

officer_roles

Role-Based Access Control (RBAC) definitions for compliance officers.

Column	Type	Constraint	Notes	Description
role_id	SERIAL	PK		Auto-incremented role ID
role_name	VARCHAR(50)	NOT NULL, UNIQUE		Role name e.g. junior_officer
can_view_alerts	BOOLEAN	DEFAULT TRUE		Permission to view alerts
can_manage_cases	BOOLEAN	DEFAULT FALSE		Permission to manage cases
can_file_sar	BOOLEAN	DEFAULT FALSE		Permission to file SARs
can_manage_rules	BOOLEAN	DEFAULT FALSE		Permission to edit compliance rules
can_manage_users	BOOLEAN	DEFAULT FALSE		Permission to manage officers
description	TEXT	Nullable		Role description

currency_rates

Daily exchange rate snapshots for multi-currency transaction normalization.

Column	Type	Constraint	Notes	Description
rate_id	SERIAL	PK		Auto-incremented ID
currency_code	CHAR(3)	NOT NULL		ISO 4217 currency code e.g. USD
currency_name	VARCHAR(50)	NOT NULL		Full currency name
rate_to_usd	DECIMAL(18,6)	NOT NULL	CHECK > 0	Exchange rate relative to USD
rate_date	DATE	NOT NULL, DEFAULT CURRENT_DATE		Date this rate applies to
		UNIQUE (currency_code, rate_date)		One rate per currency per day

Core Banking Tables

branches				
Physical bank branch locations. Ties accounts and officers to a geographic region.				
Column	Type	Constraint	Notes	Description
branch_id	SERIAL	PK		Auto-incremented branch ID
branch_name	VARCHAR(100)	NOT NULL		Branch name
city	VARCHAR(100)	NOT NULL		City of branch
country_code	CHAR(2)	FK → countries	ON DELETE RESTRICT	Branch country
address	TEXT	Nullable		Full street address
phone	VARCHAR(20)	Nullable		Branch contact number
manager_id	INT	FK → compliance_officers	ON DELETE SET NULL, Nullable	Branch manager
is_active	BOOLEAN	DEFAULT TRUE		Whether branch is operational
opened_at	DATE	NOT NULL, DEFAULT CURRENT_DATE		Branch opening date

compliance_officers				
Users of the compliance system. Receive alerts, manage cases, query the AI agent.				
Column	Type	Constraint	Notes	Description
officer_id	SERIAL	PK		Auto-incremented officer ID
role_id	INT	NOT NULL, FK → officer_roles	ON DELETE RESTRICT	Officer role
branch_id	INT	FK → branches	ON DELETE SET NULL, Nullable	Assigned branch
full_name	VARCHAR(100)	NOT NULL		Officer full name
email	VARCHAR(150)	NOT NULL, UNIQUE		Login email
phone	VARCHAR(20)	Nullable		Contact number
is_active	BOOLEAN	DEFAULT TRUE		Whether officer is active
hired_at	DATE	NOT NULL, DEFAULT CURRENT_DATE		Employment start date
last_login	TIMESTAMP	Nullable		Last system login

customers				
Bank customers. Central entity connecting accounts, cases, and risk scores.				
Column	Type	Constraint	Notes	Description
customer_id	SERIAL	PK		Auto-incremented customer ID
full_name	VARCHAR(150)	NOT NULL		Customer full name
email	VARCHAR(150)	NOT NULL, UNIQUE		Customer email
phone	VARCHAR(20)	Nullable		Contact number
nationality	CHAR(2)	FK → countries	ON DELETE SET NULL, Nullable	Nationality country code
date_of_birth	DATE	Nullable		Date of birth
occupation	VARCHAR(100)	Nullable		Stated occupation
risk_level	VARCHAR(10)	NOT NULL, DEFAULT 'low'	CHECK: low/medium/high/critical	AML risk classification
kyc_status	VARCHAR(20)	NOT NULL, DEFAULT 'pending'	CHECK: pending/verified/rejected/expired	KYC verification status
is_active	BOOLEAN	DEFAULT TRUE		Whether account is active
created_at	TIMESTAMP	DEFAULT NOW()		Registration timestamp
updated_at	TIMESTAMP	DEFAULT NOW()		Last update timestamp

accounts				
Bank accounts owned by customers, linked to a branch.				
Column	Type	Constraint	Notes	Description
account_id	SERIAL	PK		Auto-incremented account ID
customer_id	INT	NOT NULL, FK → customers	ON DELETE RESTRICT	Account owner
branch_id	INT	NOT NULL, FK → branches	ON DELETE RESTRICT	Holding branch
account_number	VARCHAR(20)	NOT NULL, UNIQUE		Unique account number
account_type	VARCHAR(20)	NOT NULL	CHECK: checking/savings/b usiness/investment	Account category
currency_code	CHAR(3)	NOT NULL, DEFAULT 'USD'		Account base currency
balance	DECIMAL(18,2)	NOT NULL, DEFAULT 0.00	CHECK >= 0	Current balance
status	VARCHAR(20)	NOT NULL, DEFAULT 'active'	CHECK: active/frozen/closed /suspended	Account status
opened_at	TIMESTAMP	DEFAULT NOW()		Account opening timestamp
closed_at	TIMESTAMP	Nullable		Account closing timestamp

transactions				
Every money movement. Core table for compliance analysis and alert generation.				
Column	Type	Constraint	Notes	Description
transaction_id	SERIAL	PK		Auto-incremented transaction ID
account_id	INT	NOT NULL, FK → accounts	ON DELETE RESTRICT	Source account
counterparty_account_id	INT	FK → accounts	ON DELETE SET NULL, Nullable	Destination account (self-ref)
transaction_type	VARCHAR(30)	NOT NULL	CHECK: deposit/withdrawal/t ransfer_domestic/transfer_int ernational/wire/cash	Transaction category
amount	DECIMAL(18,2)	NOT NULL	CHECK > 0	Original transaction amount
amount_usd	DECIMAL(18,2)	Nullable		Amount normalized to USD
currency_code	CHAR(3)	NOT NULL, DEFAULT 'USD'		Transaction currency
destination_country	CHAR(2)	FK → countries	ON DELETE SET NULL, Nullable	Destination country for intl transfers
description	TEXT	Nullable		Transaction memo
reference_number	VARCHAR(50)	UNIQUE, Nullable		External reference
status	VARCHAR(20)	NOT NULL, DEFAULT 'completed'	CHECK: pending/completed/ failed/reversed	Transaction status
is_flagged	BOOLEAN	DEFAULT FALSE		Set TRUE by trigger if alert created
created_at	TIMESTAMP	DEFAULT NOW()		Transaction timestamp

Compliance Engine Tables

compliance_rules				
The rulebook. Defines what counts as suspicious activity. Read by triggers and AI agent.				
Column	Type	Constraint	Notes	Description
rule_id	SERIAL	PK		Auto-incremented rule ID
rule_name	VARCHAR(100)	NOT NULL, UNIQUE		Human-readable rule name
rule_type	VARCHAR(30)	NOT NULL	CHECK: threshold/frequency/geography/structuring/velocity/sanctions	Rule category
threshold_amount	DECIMAL(18,2)	Nullable		Amount trigger for threshold rules
max_frequency	INT	Nullable		Max allowed transactions in window
time_window_days	INT	Nullable		Time window in days for frequency check
applies_to	VARCHAR(20)	DEFAULT 'all'	CHECK: all/international/domestic/cash	Transaction scope of rule
severity	VARCHAR(10)	NOT NULL, DEFAULT 'medium'	CHECK: low/medium/high/critical	Alert severity when rule matches
is_active	BOOLEAN	DEFAULT TRUE		Whether rule is enforced
created_at	TIMESTAMP	DEFAULT NOW()		Rule creation timestamp

alerts				
Auto-generated by trigger when a transaction matches a compliance rule.				
Column	Type	Constraint	Notes	Description
alert_id	SERIAL	PK		Auto-incremented alert ID
transaction_id	INT	NOT NULL, FK → transactions	ON DELETE CASCADE	Triggering transaction
rule_id	INT	NOT NULL, FK → compliance_rules	ON DELETE RESTRICT	Rule that was matched
assigned_to	INT	FK → compliance_officers	ON DELETE SET NULL, Nullable	Responsible officer
severity	VARCHAR(10)	NOT NULL	CHECK: low/medium/high/critical	Inherited from rule severity
status	VARCHAR(20)	NOT NULL, DEFAULT 'open'	CHECK: open/under_review/escalated/dismissed/resolved	Alert lifecycle status
is_escalated	BOOLEAN	DEFAULT FALSE		Whether escalated to a case
created_at	TIMESTAMP	DEFAULT NOW()		Alert creation timestamp
resolved_at	TIMESTAMP	Nullable		Resolution timestamp
notes	TEXT	Nullable		Officer notes

alert_comments				
Officer investigation notes added to alerts over time.				
Column	Type	Constraint	Notes	Description
comment_id	SERIAL	PK		Auto-incremented comment ID
alert_id	INT	NOT NULL, FK → alerts	ON DELETE CASCADE	Parent alert
officer_id	INT	NOT NULL, FK → compliance_officers	ON DELETE RESTRICT	Commenting officer
comment	TEXT	NOT NULL		Investigation note content
created_at	TIMESTAMP	DEFAULT NOW()		Comment timestamp

cases				
Full AML investigations. Groups multiple alerts about one customer into a workflow.				
Column	Type	Constraint	Notes	Description
case_id	SERIAL	PK		Auto-incremented case ID
customer_id	INT	NOT NULL, FK → customers	ON DELETE RESTRICT	Subject of investigation
officer_id	INT	NOT NULL, FK → compliance_officers	ON DELETE RESTRICT	Lead investigating officer
case_type	VARCHAR(20)	NOT NULL	CHECK: AML/fraud/KYC/sanctions/structuring	Investigation type
status	VARCHAR(20)	NOT NULL, DEFAULT 'open'	CHECK: open/under_review/escalated/closed_clean/closed_sar	Case lifecycle status
priority	VARCHAR(10)	NOT NULL, DEFAULT 'medium'	CHECK: low/medium/high/critical	Investigation urgency
opened_at	TIMESTAMP	DEFAULT NOW()		Case opening timestamp
closed_at	TIMESTAMP	Nullable		Case closing timestamp
summary	TEXT	Nullable		Case summary
resolution	TEXT	Nullable		Resolution details

case_alerts				
Junction table linking cases to multiple alerts (M:N relationship).				
Column	Type	Constraint	Notes	Description
case_id	INT	PK, FK → cases	ON DELETE CASCADE	Part of composite PK
alert_id	INT	PK, FK → alerts	ON DELETE CASCADE	Part of composite PK
added_at	TIMESTAMP	DEFAULT NOW()		When alert was linked to case
added_by	INT	FK → compliance_officers	ON DELETE SET NULL, Nullable	Officer who linked alert

risk_scores				
Historical risk score records per customer. Updated by system, AI agent, or officers.				
Column	Type	Constraint	Notes	Description
score_id	SERIAL	PK		Auto-incremented score ID
customer_id	INT	NOT NULL, FK → customers	ON DELETE CASCADE	Scored customer
score	DECIMAL(5,2)	NOT NULL	CHECK: 0–100	Numeric risk score
score_level	VARCHAR(10)	NOT NULL	CHECK: low/medium/high/critical	Categorized risk level
computed_by	VARCHAR(20)	NOT NULL, DEFAULT 'system'	CHECK: system/agent/officer	Who computed the score
computed_at	TIMESTAMP	DEFAULT NOW()		Computation timestamp
reasoning	TEXT	Nullable		Explanation of score factors

regulatory_reports				
Formal SAR/CTR filings submitted to financial regulators.				
Column	Type	Constraint	Notes	Description
report_id	SERIAL	PK		Auto-incremented report ID
case_id	INT	NOT NULL, FK → cases	ON DELETE RESTRICT	Associated investigation case
officer_id	INT	NOT NULL, FK → compliance_officers	ON DELETE RESTRICT	Filing officer
report_type	VARCHAR(10)	NOT NULL	CHECK: SAR/CTR/STR	Regulatory report type
regulator	VARCHAR(100)	NOT NULL		Receiving regulatory body
status	VARCHAR(20)	NOT NULL, DEFAULT 'draft'	CHECK: draft/submitted/acknowledged/rejected	Filing status
submitted_at	TIMESTAMP	Nullable		Actual submission timestamp
deadline	TIMESTAMP	Nullable		Regulatory filing deadline
reference_number	VARCHAR(100)	UNIQUE, Nullable		Regulator-assigned reference
created_at	TIMESTAMP	DEFAULT NOW()		Report creation timestamp

Screening Tables

sanctions_list				
Sanctioned individuals and entities. New customers are screened against this table.				
Column	Type	Constraint	Notes	Description
sanction_id	SERIAL	PK		Auto-incremented sanction ID
full_name	VARCHAR(200)	NOT NULL		Sanctioned entity name
entity_type	VARCHAR(20)	NOT NULL	CHECK: individual/organization/vessel/aircraft	Type of sanctioned entity
country_code	CHAR(2)	FK → countries	ON DELETE SET NULL, Nullable	Country of entity
sanction_type	VARCHAR(50)	Nullable		e.g. OFAC, UN, EU
listed_by	VARCHAR(100)	NOT NULL		Sanctioning authority
listed_at	DATE	NOT NULL		Date entity was listed
is_active	BOOLEAN	DEFAULT TRUE		Whether sanction is current
notes	TEXT	Nullable		Additional details

pep_list				
Politically Exposed Persons. Customers matched here receive enhanced monitoring.				
Column	Type	Constraint	Notes	Description
pep_id	SERIAL	PK		Auto-incremented PEP ID
full_name	VARCHAR(200)	NOT NULL		PEP full name
position	VARCHAR(200)	NOT NULL		Political position held
country_code	CHAR(2)	FK → countries	ON DELETE SET NULL, Nullable	Country of political position
pep_level	VARCHAR(20)	NOT NULL	CHECK: domestic/foreign/international	PEP classification level
is_active	BOOLEAN	DEFAULT TRUE		Whether still in position
since_date	DATE	Nullable		Date position was assumed
notes	TEXT	Nullable		Additional context

Analytics Tables

transaction_patterns				
Aggregated behavioral statistics per customer used for anomaly detection.				
Column	Type	Constraint	Notes	Description
pattern_id	SERIAL	PK		Auto-incremented pattern ID
customer_id	INT	NOT NULL, FK → customers	ON DELETE CASCADE	Customer being profiled
avg_transaction	DECIMAL(18,2)	Nullable		Average transaction amount (USD)
max_transaction	DECIMAL(18,2)	Nullable		Largest single transaction (USD)
monthly_volume	DECIMAL(18,2)	Nullable		Average monthly transaction volume
typical_countries	TEXT	Nullable		Comma-separated common destination countries
international_pct	DECIMAL(5,2)	Nullable		% of transactions that are international
cash_pct	DECIMAL(5,2)	Nullable		% of transactions that are cash
computed_at	TIMESTAMP	DEFAULT NOW()		When pattern was last computed

monthly_reports				
Aggregated monthly compliance stats per branch. Used for OLAP and rollup queries.				
Column	Type	Constraint	Notes	Description
report_id	SERIAL	PK		Auto-incremented report ID
branch_id	INT	NOT NULL, FK → branches	ON DELETE CASCADE	Reporting branch
report_month	DATE	NOT NULL		Month (first day of month)
total_transactions	INT	DEFAULT 0		Transaction count for month
total_volume_usd	DECIMAL(18,2)	DEFAULT 0		Total transaction volume (USD)
total_alerts	INT	DEFAULT 0		Alerts generated that month
total_cases	INT	DEFAULT 0		Cases opened that month
sars_filed	INT	DEFAULT 0		SAR filings submitted
generated_at	TIMESTAMP	DEFAULT NOW()		When report was generated
		UNIQUE (branch_id, report_month)		One report per branch per month

audit_log				
Immutable record of every officer action. Required for regulatory examination.				
Column	Type	Constraint	Notes	Description
log_id	SERIAL	PK		Auto-incremented log ID
officer_id	INT	NOT NULL, FK → compliance_officers	ON DELETE RESTRICT	Acting officer
case_id	INT	FK → cases	ON DELETE SET NULL, Nullable	Related case if applicable
alert_id	INT	FK → alerts	ON DELETE SET NULL, Nullable	Related alert if applicable
action	VARCHAR(100)	NOT NULL		Action performed e.g. status_change
entity_type	VARCHAR(50)	Nullable		Type of entity acted upon
entity_id	INT	Nullable		ID of entity acted upon
details	TEXT	Nullable		Full action description
ip_address	INET	Nullable		Officer IP address
action_at	TIMESTAMP	DEFAULT NOW()		Timestamp of action